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(57) Abstract :

This invention concerns an Artificial Intelligence (AI)-based Industrial Internet of Things (IIoT) system designed for the real-time monitoring, condition evaluation, anomaly detection, and predictive maintenance of industrial machinery. The system includes a number of industrial sensors which are set up to collect operating parameters such as vibration, temperature, pressure, current, voltage, rotational speed, acoustic signals and other parameters that are specific to the equipment; an IIoT edge gateway that is capable of acquiring, pre-processing, normalising and securely transmitting the sensor data; an edge intelligence module that carries out real-time feature extraction and initial anomaly detection; a communication module that sends the processed data to an industrial data platform; an AI/ML analytics engine that is able to identify abnormal operating conditions, classify the states of the equipment, estimate its health and predict possible faults; and a visualization and decision-support module that produces equipment-health indicators, alerts, maintenance recommendations and historical analytics. The framework also includes continuous feedback from the equipment's condition to the AI/ML analytics engine so that the models can be adaptively updated. The framework in question decreases the need for manual inspection, allows for the early detection of abnormalities in the equipment, supports predictive maintenance, enhances the availability of the equipment and promotes intelligent industrial decision-making.

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